UNIFORM SHEAFY TATE DOMAINS THAT ARE NOT STABLY UNIFORM

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Abstract. Work of Buzzard–Verberkmoes and, independently, Mihara shows that stable uniformity is a sufficient criterion for a Tate Huber ring to be sheafy, meaning that the structure presheaf of topological rings on its associated adic space is a sheaf. In this paper we construct examples of uniform sheafy rings which are not stably uniform, answering Question 7 of Kedlaya's *Nonarchimedean Scottish Book*. The examples were found in collaboration with ChatGPT 5.6 Sol. The first has been fully formalised in Lean 4; its proof rests on showing that a particular strict pullback of complete Tate rings is preserved by rational localisation. Formalisation of the second is ongoing: the construction, uniformity, sheafiness, and the nonuniform domain chart have been checked, but reducedness of every finite iterated rational localisation has not.

1. Introduction

The point at issue is the behaviour of rational localisations. The examples of Buzzard–Verberkmoes and Mihara show that a uniform Tate Huber ring can lose uniformity on a rational chart and, separately, that uniformity alone does not imply sheafiness [5, 15]. They leave open whether sheafiness itself prevents this loss of uniformity. Hansen and Kedlaya ask this explicitly [7, Remark 3.16]; it is also Question 7 of Kedlaya's *Nonarchimedean Scottish Book* [16]. The answer is no.

We begin with the finite-jet pullback. This is a nonnoetherian uniform integral domain for which a distinguished rational localisation is nonreduced, and hence is no longer uniform. Let k be a complete discretely valued nonarchimedean field, with valuation ring k°, uniformizer ϖ, and residue field $\widetilde{k}$. Put

$$L_0 = k^\circ\langle W, W^{-1}\rangle, \qquad B_0 = k^\circ\langle W, Q\rangle/(Q^2),$$
$$C_0 = L_0\langle Q\rangle, \qquad D_0 = L_0\langle Q\rangle/(Q^2),$$

where $k^\circ\langle W, W^{-1}\rangle$ means $k^\circ\langle W, V\rangle/(WV-1)$. There are natural maps $B_0 \to D_0$ and $C_0 \twoheadrightarrow D_0$. Define

$$A_0 = B_0 \times_{D_0} C_0, \qquad A = A_0[1/\varpi],$$

and endow A with the topology defined by the ring of definition A_0.

The main result is the following.

Theorem 1.1. *The ring A is a complete uniform nonnoetherian Tate k-algebra and an integral domain, with $A^\circ = A_0$. The Huber pair (A, A°) is sheafy. However,*

$$A\left\langle \frac{W}{\varpi} \right\rangle \cong k\langle X, Q\rangle/(Q^2), \qquad X = \frac{W}{\varpi},$$

so A is not stably uniform.

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The proof uses Milnor squares of complete Tate rings. Recall that a cartesian square of rings with a surjective leg is called a Milnor square [14, Section 2]; the complete Tate setting is used by Kerz–Saito–Tamme in their work on analytic K-theory [11, Section 3].

Definition 1.2. A commutative square of complete Tate k-algebras

$$\begin{array}{ccc} R & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

is a *strict Milnor square* if it is cartesian, the sequence of underlying topological k-vector spaces

$$0 \longrightarrow R \longrightarrow B \oplus C \xrightarrow{(b,c)\mapsto \bar{b}-\bar{c}} D \longrightarrow 0$$

is strict exact, and $C \to D$ is a strict surjection.

After rotating the square if necessary, the initial Milnor square is strict by the nonarchimedean open mapping theorem [11, Lemma 3.1]. Our main input is that strictness persists after completed rational localisation and is compatible with refinement. This allows sheafiness to be transferred from B, C, D to R. The associated square of affinoid adic spaces is a pushout among affinoid spaces, but not among all adic spaces; see remark 6.3. The second example shows that the failure of stable uniformity need not come from a nonreduced localisation.

Both examples were found in collaboration with ChatGPT 5.6 Sol. The first has been formalised in Lean, and substantial parts of the second have also been checked. Appendix A gives the precise scope of the two developments and relates their statements to the definitions used here.

Remark 1.3 (Relation with earlier work)**.** Buzzard–Verberkmoes construct a uniform affinoid ring with a nonuniform rational localisation and, separately, a uniform affinoid ring whose structure presheaf is not a sheaf [5, Propositions 17 and 18]. Mihara gives further examples of both phenomena [15]. These examples do not give a uniform sheafy ring which is not stably uniform.

The strict Koszul result used in lemma 5.1 is due to Ben-Bassat–Kremnizer and Bambozzi–Kremnizer [2, Definition 5.7 and Lemmas 5.13–5.14] [3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13]. We use its lattice form to prove that rational localisation preserves the finite-jet Milnor square. The resulting uniform sheafy domain which is not stably uniform appears to be new.

The weighted-parity construction of section 7 is closer to the infinite monomial-support examples of Buzzard–Verberkmoes and Mihara. Its additional feature is that rational localisation can be calculated in a noetherian finite head.

2. Conventions and notation

We follow Huber for Huber pairs, adic spectra, rational subsets, and rational localisation [8], and use Wedhorn's account for strong noetherianity and sheafiness [18, Proposition and Definition 6.36, Section 8.1]; the bounded-denominator language for strictness follows Buzzard–Verberkmoes [5], and the quotient presentation of a rational localisation is also discussed by Hansen–Kedlaya [7].

- *Tate topology.* Throughout, E denotes a complete Tate k-algebra. In the discretely valued setting of this paper, this means that E contains a topologically nilpotent unit and admits an open bounded k°-subalgebra E_0 such that

$$E = E_0[1/\varpi], \qquad \{\varpi^n E_0\}_{n\geq 0}$$

is a basis of neighbourhoods of zero. Such an E_0 is called a *ring of definition*. We always choose it closed, hence ϖ-adically complete. A subset of E is bounded if it is contained in $\varpi^{-r}E_0$ for some $r \geq 0$. If M is a complete topological k-vector space, an open bounded k°-submodule $M_0 \subset M$ will be called a lattice.

- *Uniformity and noetherianity.* We write E° for the ring of power-bounded elements. The ring E is *uniform* if E° is bounded, and *strongly noetherian* if $E\langle Z_1, \dots, Z_s\rangle$ is noetherian for every $s \geq 0$. A Tate ring is *stably uniform* if every rational localisation is uniform.
- *Huber pairs and sheafiness.* A *ring of integral elements* is an open subring $E^+ \subseteq E^\circ$ which is integrally closed in E. We call the Huber pair (E, E^+) *sheafy* when its structure presheaf is a sheaf of complete topological rings; in particular, the map to the matching-family equaliser for a finite rational cover is a homeomorphism.
- *Strictness.* We use the following bounded-denominator form of strictness. Let $u : M \to N$ be a continuous k-linear map and choose lattices $M_0 \subset M$ and $N_0 \subset N$. Then u is strict if and only if there exists $a \geq 0$ such that

$$\varpi^a\big(u(M) \cap N_0\big) \subset u(M_0). \tag{1}$$

For a surjection this becomes $\varpi^a N_0 \subset u(M_0)$. For an injection it says that the lattice induced from the target is bounded in the source. This is the criterion used by Buzzard–Verberkmoes [5, Lemma 2]. We use the term *strict Milnor square* in the sense of definition 1.2; the underlying algebraic terminology is standard [14, 11].
- *Rational localisation.* We take every rational datum to have at least one numerator; an empty list may be padded by $f_1 = 0$. Let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum in E, so the ideal generated by $g, f_1, \dots, f_m$ is open. Since E is a Tate k-algebra, this ideal is the unit ideal. After multiplying the entire datum by a common power of ϖ, we may assume that all entries lie in E_0.

The rational localisation E_α is the separated ϖ-adic completion of $E[1/g]$ for the ring of definition generated by

$$E_0\left[\frac{f_1}{g}, \dots, \frac{f_m}{g}\right].$$

It is independent of the ring of definition and of the presentation of the rational domain, and it is transitive under rational refinement. Put

$$T_E = E\langle T_1, \dots, T_m\rangle = E_0\langle T_1, \dots, T_m\rangle[1/\varpi], \qquad r_i = gT_i - f_i,$$

and

$$I_{E,\alpha} := (r_1, \dots, r_m) = \operatorname{im}\left(T_E^m \xrightarrow{d_{1,E}} T_E\right) \subset T_E, \qquad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i(gT_i - f_i), \tag{2}$$

Then

$$E_\alpha \cong T_E/\overline{I_{E,\alpha}}. \tag{3}$$

Here the bar is necessary until the ideal has been shown to be closed. This is the standard quotient presentation of rational localisation; see [8, (1.2), Proposition 1.3, Lemma 1.5] and [7, Definition 3.6].

3. The global ring

In this section we identify the finite-jet algebra with a concrete subring of C and prove that it is a complete uniform domain. Recall that

$$L_0 = k^\circ\langle W, W^{-1}\rangle, \qquad B_0 = k^\circ\langle W, Q\rangle/(Q^2),$$
$$C_0 = L_0\langle Q\rangle, \qquad D_0 = L_0\langle Q\rangle/(Q^2),$$

where $k^\circ\langle W, W^{-1}\rangle$ denotes $k^\circ\langle W, V\rangle/(WV-1)$. The natural maps $B_0 \to D_0$ and $C_0 \twoheadrightarrow D_0$ define

$$A_0 = B_0 \times_{D_0} C_0.$$

We write

$$A = A_0[1/\varpi], \quad L = L_0[1/\varpi], \quad B = B_0[1/\varpi], \quad C = C_0[1/\varpi], \quad D = D_0[1/\varpi].$$

Each of these Tate rings carries the topology defined by its subscript-zero ring.

The map $C_0 \to D_0$ has a continuous k°-linear section given by truncation in the Q-variable:

$$f_0 + Qf_1 \longmapsto f_0 + Qf_1.$$

Consequently

$$0 \longrightarrow A_0 \longrightarrow B_0 \oplus C_0 \longrightarrow D_0 \longrightarrow 0 \tag{4}$$

is exact, and A_0 is ϖ-adically complete. After inverting ϖ we obtain a strict Milnor square

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D. \end{array}$$

In fact, the chosen rings of definition already make the integral row exact, so no powers of ϖ are lost in the defining square.

The map $B_0 \to D_0$ is injective. Projection to C_0 therefore identifies A_0 with the subring

$$A_0 = \left\{ f_0(W) + Qf_1(W) + Q^2 h(W, W^{-1}, Q) : f_0, f_1 \in k^\circ\langle W\rangle,\ h \in C_0 \right\}. \tag{5}$$

Equivalently, A_0 consists of the restricted series

$$\sum_{(a,b)\in S} c_{a,b} W^a Q^b, \qquad S = \{(a,b) \in \mathbf{Z} \times \mathbf{N} : b \le 1 \Rightarrow a \ge 0\}, \tag{6}$$

with $c_{a,b} \in k^\circ$. Here restricted means that, for every n, only finitely many coefficients are nonzero modulo ϖ^n. The set S is closed under addition, so this is a subring of C_0. The same description with coefficients in k gives $A \subset C$.

Equip A, B, C, D with the coefficientwise Gauss norms whose unit balls are the displayed rings of definition. Write

$$\iota_B : A \to B, \qquad \iota_C : A \to C, \qquad \rho_B : B \to D, \qquad \rho_C : C \to D$$

for the structural maps, and give A the restricted Gauss norm from C. For $a = f_0 + Qf_1 + Q^2 h$, the coefficient formulas give

$$\|\iota_C(a)\|_C = \|a\|_A, \qquad \|\iota_B(a)\|_B \le \|a\|_A, \qquad \|\rho_B(b)\|_D \le \|b\|_B, \qquad \|\rho_C(c)\|_D \le \|c\|_C. \tag{7}$$

Thus every structural map is norm-nonincreasing.

Proposition 3.1. *The ring A is a complete uniform nonnoetherian Tate k-algebra and an integral domain. Moreover,*

$$A^\circ = A_0.$$

Proof. For $0 \neq c \in C$, define

$$\nu(c) = \max\{n \in \mathbf{Z} : c \in \varpi^n C_0\}.$$

This maximum exists because C_0 is ϖ-adically separated and $C = C_0[1/\varpi]$. We claim that

$$\nu(cc') = \nu(c) + \nu(c') \qquad (c, c' \neq 0). \tag{8}$$

After multiplying c and c' by suitable powers of ϖ, it is enough to treat the case $c, c' \in C_0 \setminus \varpi C_0$. Their reductions are nonzero elements of

$$C_0/\varpi C_0 \cong \widetilde{k}[W, W^{-1}, Q].$$

This is a domain, so the product of the reductions is nonzero. This proves (8); in particular, C is a domain.

By the preceding coefficient description, A consists of the same restricted series as A_0, but with coefficients in k. Such a series lies in C_0 precisely when all of its coefficients lie in k°. Hence

$$A_0 = A \cap C_0.$$

Every element of A_0 is power-bounded because A_0 is a subring. Conversely, let $a \in A \setminus A_0$. Then $\nu(a) < 0$, and by (8), $\nu(a^n) = n\nu(a)$. Hence the powers of a cannot all lie in any fixed $\varpi^{-r}A_0$. Thus a is not power-bounded. It follows that $A^\circ = A_0$, so A is uniform. Completeness follows from the completeness of A_0, and A is a domain because it is a subring of C.

For nonnoetherianity, let $J = \ker(\iota_B) = Q^2 C \subset A$. If J were generated by $a_1, \dots, a_r$, then, for every $\ell \in L$, comparison of the Q^2-coefficient in an expression

$$Q^2 \ell = \sum_i x_i a_i, \qquad x_i \in A,$$

would express ℓ as a $k\langle W\rangle$-linear combination of the Q^2-coefficients of the a_i. This would make $L = k\langle W, W^{-1}\rangle$ a finite $k\langle W\rangle$-module, which is impossible: a finite algebra is integral, whereas a monic equation for W^{-1}, after multiplication by a power of W, would put 1 in the proper ideal $(W) \subset k\langle W\rangle$. Thus J is not finitely generated, and A is not noetherian. □

4. A nonuniform rational localisation

We next compute the rational chart which witnesses the failure of stable uniformity.

Consider the datum

$$\alpha_{\text{in}} = (W; \varpi).$$

The ideal (ϖ, W) is open, since ϖ is a unit in A. Let

$$A_{\text{in}} = A\left\langle \frac{W}{\varpi} \right\rangle$$

be the corresponding rational localisation on $|W| \leq |\varpi|$, and write $X = W/\varpi$. By the ring-of-definition construction, A_{in} is the separated completion of the subring generated by A_0 and X.

Proposition 4.1. *There is a canonical isomorphism of complete topological rings*

$$A_{\text{in}} \cong k\langle X, Q\rangle/(Q^2), \qquad W \longmapsto \varpi X.$$

Proof. Let

$$q\colon A_0[X] \longrightarrow A[X]/(\varpi X - W)$$

be the composite of the inclusion $A_0[X] \hookrightarrow A[X]$ with the quotient map, and set

$$G_0 := q(A_0[X]).$$

Since ϖ is a unit in A, evaluation at $X = W/\varpi$ gives a canonical isomorphism

$$A[X]/(\varpi X - W) \xrightarrow{\sim} A.$$

Under this identification,

$$G_0 = A_0[W/\varpi] \subset A;$$

it is the ring of definition whose separated ϖ-adic completion defines A_{in}. If $y \in Q^2C_0$, its constant and linear Q-coefficients vanish, so $W^{-n}y \in A_0$ for every $n \geq 0$. In G_0 one has

$$y = W^n(W^{-n}y) = \varpi^n X^n(W^{-n}y) \in \varpi^n G_0. \tag{9}$$

Thus Q^2C_0 maps to zero in the separated completion. After multiplying by a suitable power of ϖ, the same conclusion holds for every element of Q^2C.

Put

$$T_0 = k^\circ\langle X, Q\rangle/(Q^2), \qquad T = T_0[1/\varpi].$$

Using the unique expression in (5), define

$$f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(\varpi X) + Qf_1(\varpi X).$$

This sends A_0 into T_0, and it sends W/ϖ to X. Hence it extends to a continuous homomorphism $A_{\mathrm{in}} \to T$.

Conversely, X and Q belong to the ring of definition of A_{in}, and (9) gives $Q^2 = 0$ there. They therefore define a continuous homomorphism $T \to A_{\mathrm{in}}$. The two composites fix X and Q. On the dense image of G_0, the second composite differs from the identity only by an element of Q^2C_0, which is zero by (9). The two maps are therefore inverse. □

Proposition 4.2. *The ring A is not stably uniform.*

Proof. In $T = k\langle X, Q\rangle/(Q^2)$, every element of the line kQ is nilpotent and hence power-bounded. This line is not bounded with respect to the ring of definition T_0: for every $r \geq 0$,

$$\varpi^r\left(\varpi^{-(r+1)}Q\right) = \varpi^{-1}Q \notin T_0.$$

Thus T is not uniform. By proposition 4.1, a rational localisation of the uniform ring A is nonuniform. □

5. Localisation of the Milnor square

In this section we prove that rational localisation preserves the defining pullback. We first record the strict Koszul estimate needed for the proof.

5.1. **Koszul complexes for rational localisations.** Let E be an affinoid k-algebra, choose a noetherian ϖ-adically complete ring of definition $E_0 \subset E$, and let $f_1, \ldots, f_m, g \in E_0$ generate the unit ideal in E. Put

$$T_E = E\langle T_1, \ldots, T_m\rangle, \qquad T_{E,0} = E_0\langle T_1, \ldots, T_m\rangle,$$

and set $r_i = gT_i - f_i$.

We give T_E the topology for which $\{\varpi^n T_{E,0}\}_{n\geq 0}$ is a basis of neighbourhoods of zero. The term $\bigwedge^q T_E^m$ has the finite-module topology defined by the lattice $\bigwedge^q T_{E,0}^m$; equivalently, in the standard exterior basis it has the finite product topology. Images have the subspace topology. Thus a differential is strict when it is open onto its image, or equivalently when its coimage is homeomorphic to its image.

The following strict Koszul result is known [2, Definition 5.7 and Lemmas 5.13–5.14] [3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13]. We state the two bounded-denominator consequences which will be used in the pullback argument.

Lemma 5.1. *The Koszul complex*

$$0 \longrightarrow \bigwedge^m T_E^m \longrightarrow \cdots \longrightarrow \bigwedge^2 T_E^m \xrightarrow{d_{2,E}} T_E^m \xrightarrow{d_{1,E}} T_E$$

is exact in positive degrees. Every differential is continuous and strict onto its image, and every image is closed. In particular, if

$$I_E = (r_1, \ldots, r_m) = \operatorname{im}(d_{1,E}) \subset T_E, \qquad d_{1,E}(u_1, \ldots, u_m) = \sum_i u_i r_i,$$

then I_E is closed. Moreover, there exists $h_E \in \mathbf{Z}_{\geq 0}$ such that

$$\varpi^{h_E}(I_E \cap T_{E,0}) \subset d_{1,E}(T_{E,0}^m). \tag{10}$$

There also exists $z_E \in \mathbf{Z}_{\geq 0}$ such that

$$\varpi^{z_E}\big(\ker(d_{1,E}) \cap T_{E,0}^m\big) \subset d_{2,E}\big(\textstyle\bigwedge^2 T_{E,0}^m\big). \tag{11}$$

Proof. Positive-degree exactness follows from the strict Koszul regularity result cited above. Since T_E is a complete Tate ring with noetherian ring of definition $T_{E,0}$, Huber's finite-module lemmas show that the differentials are continuous and strict and that their images are closed [8, Lemmas 2.3 and 2.4]. Apply (1) to $d_{1,E} : T_E^m \to I_E$, using the target lattice $I_E \cap T_{E,0}$. Exactness identifies $\operatorname{im}(d_{2,E}) = \ker(d_{1,E})$; applying the same criterion to $d_{2,E}$ with target lattice $\ker(d_{1,E}) \cap T_{E,0}^m$ gives (10) and (11). □

5.2. **The defining ideals under pullback.** Fix a rational datum

$$\alpha = (f_1, \ldots, f_m; g)$$

in A. Multiplying all entries by a common power of ϖ, we assume $f_1, \ldots, f_m, g \in A_0$. For $E \in \{A, B, C, D\}$ use the notation T_E, $T_{E,0}$, and $d_{1,E}$ introduced above, and write

$$I_E = (gT_1 - f_1, \ldots, gT_m - f_m) \subset T_E.$$

The exact integral row (4) and the coefficientwise section of $C_0 \to D_0$ give an exact sequence

$$0 \longrightarrow T_{A,0} \longrightarrow T_{B,0} \oplus T_{C,0} \longrightarrow T_{D,0} \longrightarrow 0. \tag{12}$$

The map $T_{C,0} \to T_{D,0}$ is surjective, as are the induced maps on finite free modules and their exterior powers. Exactness is coefficientwise: a compatible restricted coefficient pair lifts to A_0, and the lifted coefficients still tend to zero ϖ-adically.

Lemma 5.2. *The ideal I_A is closed in T_A, and the canonical map*

$$I_A \longrightarrow I_B \times_{I_D} I_C$$

is an isomorphism. Moreover,

$$0 \longrightarrow I_A \longrightarrow I_B \oplus I_C \longrightarrow I_D \longrightarrow 0 \tag{13}$$

is strict exact.

Proof. We first prove the algebraic statement. Let $(x_B, x_C) \in I_B \times_{I_D} I_C$. Choose $u_B \in T_B^m$ and $u_C \in T_C^m$ with $d_{1,B}(u_B) = x_B$ and $d_{1,C}(u_C) = x_C$. Their difference in T_D^m lies in $\ker(d_{1,D})$. By lemma 5.1, it is $d_{2,D}(v_D)$ for some $v_D \in \bigwedge^2 T_D^m$. Lift v_D to $v_C \in \bigwedge^2 T_C^m$ and replace u_C by $u_C + d_{2,C}(v_C)$. The new pair of coefficient vectors has the same image in T_D^m, hence comes from $u_A \in T_A^m$ by (12). Then $d_{1,A}(u_A)$ maps to (x_B, x_C). This proves the asserted algebraic isomorphism.

We now make this correction with bounded denominators. The exponents h_B and h_C make the coefficient lifts integral, whilst z_D makes the syzygy correction integral. Put $h = \max\{h_B, h_C\}$, and suppose that (x_B, x_C) belongs to the pullback and lies in $T_{B,0} \oplus T_{C,0}$. Choose $u_B \in T^m_{B,0}$ and $u_C \in T^m_{C,0}$ such that

$$d_{1,B}(u_B) = \varpi^h x_B, \qquad d_{1,C}(u_C) = \varpi^h x_C.$$

The difference $s = (u_B)_D - (u_C)_D$ belongs to $\ker(d_{1,D}) \cap T^m_{D,0}$. By (11), choose $v_D \in \bigwedge^2 T^m_{D,0}$ with $d_{2,D}(v_D) = \varpi^{z_D} s$. Lift v_D to $v_C \in \bigwedge^2 T^m_{C,0}$ and set

$$u'_C = \varpi^{z_D} u_C + d_{2,C}(v_C).$$

Then $(u'_C)_D = \varpi^{z_D}(u_B)_D$ and $d_{1,C}(u'_C) = \varpi^{h+z_D} x_C$. By (12), the compatible integral pair $(\varpi^{z_D} u_B, u'_C)$ comes from $u_A \in T^m_{A,0}$. Thus

$$\varpi^{h+z_D}\big((I_B \times_{I_D} I_C) \cap (T_{B,0} \oplus T_{C,0})\big) \subset d_{1,A}(T^m_{A,0}). \tag{14}$$

This proves strictness of the pullback identification. The fibre product $I_B \times_{I_D} I_C$ is closed in $T_B \oplus T_C$, and (12) identifies T_A strictly with the kernel of $T_B \oplus T_C \to T_D$. It follows from (14) that I_A is closed in T_A.

Finally, the map $I_B \oplus I_C \to I_D$ is surjective: lift a coefficient vector in T^m_D to T^m_C and apply d_1. If the element of I_D is integral, then (10) for $E = D$ shows that a fixed power of ϖ has an integral coefficient vector, which can be lifted to $T^m_{C,0}$. Thus this surjection is strict. Together with (14), this proves strict exactness of (13). □

Proposition 5.3. *For every rational datum α in A, the canonical sequence*

$$0 \longrightarrow A_\alpha \longrightarrow B_\alpha \oplus C_\alpha \longrightarrow D_\alpha \longrightarrow 0 \tag{15}$$

is strict exact, and $C_\alpha \to D_\alpha$ is a strict surjection. Equivalently,

$$A_\alpha \cong B_\alpha \times_{D_\alpha} C_\alpha.$$

The coefficient maps used in these identifications commute with rational restriction.

Proof. By lemma 5.2, all four ideals are closed. Hence (3) gives

$$E_\alpha = T_E / I_E \qquad (E = A, B, C, D).$$

The two exact rows (12) and (13) give algebraic exactness of (15).

Let $E_{\alpha,0}$ be the image of $T_{E,0}$ in E_α; this is a ring of definition. Since $T_{C,0} \to T_{D,0}$ is surjective, so is $C_{\alpha,0} \to D_{\alpha,0}$. Thus $C_\alpha \to D_\alpha$ is a strict surjection.

It remains to check strictness at the left. Let $x \in A_\alpha$ have images in $B_{\alpha,0}$ and $C_{\alpha,0}$. Choose representatives $b_0 \in T_{B,0}$ and $c_0 \in T_{C,0}$. Their difference

$$\delta = (b_0)_D - (c_0)_D$$

belongs to $I_D \cap T_{D,0}$. Choose h_D as in (10). There is $u_D \in T^m_{D,0}$ with $d_{1,D}(u_D) = \varpi^{h_D}\delta$. Lift u_D to $u_C \in T^m_{C,0}$. Then

$$\Big(\varpi^{h_D} b_0, \varpi^{h_D} c_0 + d_{1,C}(u_C)\Big)$$

is a compatible pair in $T_{B,0} \oplus T_{C,0}$, so it comes from $T_{A,0}$. Its class in A_α is $\varpi^{h_D} x$. Therefore

$$\varpi^{h_D}\Big(A_\alpha \cap (B_{\alpha,0} \oplus C_{\alpha,0})\Big) \subset A_{\alpha,0},$$

which is the bounded-denominator criterion for strictness.

All maps used above come from the coefficient maps $A \to B, C$ and $B, C \to D$. More generally, a homomorphism $E \to E'$ carrying a rational datum α to α' induces a coefficientwise map $T_E \to T_{E'}$ which sends the relations for α to those for α'. The pullback identifications therefore commute with these maps. Applying this to rational restriction, and then using presentation independence and transitivity, proves the asserted compatibility under refinement. □

6. Sheafiness

We isolate the elementary descent statement used to transfer sheafiness from the three comparison rings.

Lemma 6.1 (Sheaf transfer)**.** *Let*

$$\begin{array}{ccc} R & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

be a commutative square of complete Tate Huber pairs. For $E \in \{B, C, D\}$, write

$$p_E : X_E = \mathrm{Spa}(E, E^+) \longrightarrow X_R = \mathrm{Spa}(R, R^+)$$

for the induced map. For a rational domain $U \subset X_R$, put $U_E = p_E^{-1}(U)$, and set $U_R = U$. Suppose that for every rational domain $U \subset X_R$ the section rings satisfy a natural strict exact sequence

$$0 \longrightarrow \mathcal{O}_R(U) \longrightarrow \mathcal{O}_B(U_B) \oplus \mathcal{O}_C(U_C) \longrightarrow \mathcal{O}_D(U_D) \longrightarrow 0. \tag{16}$$

If the Huber pairs associated with B, C, D are sheafy as complete topological rings, then the Huber pair associated with R is sheafy as a complete topological ring.

Proof. We first work on the rational basis. Let $U = \bigcup_{i=1}^n U_i$ be a finite rational covering. For $E \in \{R, B, C, D\}$, let

$$r_E : \mathcal{O}_E(U_E) \longrightarrow \prod_i \mathcal{O}_E((U_i)_E)$$

denote product restriction, using $U_R = U$ and $(U_i)_R = U_i$ as specified in the statement. The first map in (16) is a topological embedding, and the same is true after replacing U by any U_i. Since B and C are sheafy, r_B and r_C are topological embeddings. Naturality gives a commutative square

$$\begin{array}{ccc} \mathcal{O}_R(U) & \longrightarrow & \mathcal{O}_B(U_B) \oplus \mathcal{O}_C(U_C) \\ \downarrow r_R & & \downarrow r_B \oplus r_C \\ \prod_i \mathcal{O}_R(U_i) & \longrightarrow & \prod_i \big(\mathcal{O}_B((U_i)_B) \oplus \mathcal{O}_C((U_i)_C)\big). \end{array}$$

The top and right arrows are topological embeddings, as is the bottom arrow. It follows that r_R is a topological embedding.

Now let $x_i \in \mathcal{O}_R(U_i)$ be a matching family. Its images in the B- and C-section rings glue to elements

$$b \in \mathcal{O}_B(U_B), \qquad c \in \mathcal{O}_C(U_C).$$

Their images in $\mathcal{O}_D(U_D)$ agree after restriction to every $(U_i)_D$. Since the structure presheaf for D is separated, they agree globally. Exactness on U therefore gives $x \in \mathcal{O}_R(U)$ mapping to (b, c). Exactness on each U_i shows that $x|_{U_i} = x_i$, and uniqueness follows in the same way. Thus r_R identifies $\mathcal{O}_R(U)$ homeomorphically with the matching-family equalizer, which is closed in the finite product because all section rings are Hausdorff.

The underlying presheaf is therefore a sheaf by the sheaf-on-a-basis theorem [17, Tag 009O]. To retain the topology on arbitrary opens, let $V = \bigcup_\lambda V_\lambda$ be an open covering and let M be its matching-family equalizer with the subspace topology. For every rational $U \subset V$, quasi-compactness gives a finite rational cover of U subordinate to the V_λ. Restriction from M to that finite cover, followed by the rational-basis result, gives a continuous map $M \to \mathcal{O}_R(U)$, independent of the chosen subordinate cover. These maps are compatible under rational restriction. Since

$$\mathcal{O}_R(V) = \varprojlim_{U \subset V,\ U \text{ rational}} \mathcal{O}_R(U)$$

with its projective-limit topology, the inverse of the algebraic gluing map $\mathcal{O}_R(V) \to M$ is continuous. Hence the gluing map is a homeomorphism, as required. □

Theorem 6.2. *The Huber pair (A, A°) is sheafy.*

Proof. Give B, C, D their maximal rings of integral elements. The max/Gauss estimates (7) show that every structural map is norm-nonincreasing. Consequently it sends bounded subsets to bounded subsets; in particular, if x is power-bounded, then the set of powers of its image is the image of the bounded set $\{x^n : n \geq 0\}$. The structural maps therefore send power-bounded elements to power-bounded elements and define morphisms of Huber pairs.

For a rational domain $U \subset \operatorname{Spa}(A, A^\circ)$, its inverse image in each comparison spectrum is defined by the image of the same rational datum. By proposition 5.3, the corresponding section rings form the natural exact sequence required in lemma 6.1. Each of B, C, D is an affinoid k-algebra, hence is sheafy by Huber's theorem [8, Theorem 2.2]. The conclusion follows from lemma 6.1. □

Remark 6.3 (Affinoid versus global pushouts)**.** The coefficient descriptions of the maximal plus rings give

$$A^\circ = B^\circ \times_{D^\circ} C^\circ.$$

Indeed, a compatible pair which is power-bounded in B and C has its constant and linear Q-coefficients in both $k\langle W\rangle$ and L_0; their intersection is $k^\circ\langle W\rangle$, while the higher Q-tail is already integral in C_0. Together with the pullback topology, this says that the defining square is a pullback of complete Huber pairs.

Put $X_E = \operatorname{Spa}(E, E^\circ)$ for $E \in \{A, B, C, D\}$. Huber's affinoid mapping theorem identifies morphisms between affinoid adic spaces with continuous morphisms of the corresponding complete Huber pairs [8, Proposition 2.1]. Hence, for every affinoid adic space Z, the natural map

$$\operatorname{Hom}(X_A, Z) \longrightarrow \operatorname{Hom}(X_B, Z) \times_{\operatorname{Hom}(X_D, Z)} \operatorname{Hom}(X_C, Z)$$

is bijective. Thus the contravariant square

$$\begin{array}{ccc} X_D & \longrightarrow & X_C \\ \downarrow & & \downarrow \\ X_B & \longrightarrow & X_A \end{array}$$

is a pushout in the full subcategory of affinoid adic spaces.

This does not make it a pushout in the category of all adic spaces: there the same map must be bijective for an arbitrary, possibly nonaffinoid, target Z. In fact, uniqueness already fails for an explicit nonaffinoid target. Let

$$U_1 = \{x \in X_A : |Q^2(x)| \leq |\varpi(x)|\}, \qquad U_2 = \{x \in X_A : |1| \leq |W(x)|\}, \qquad U = U_1 \cup U_2.$$

The map $X_B \to X_A$ factors through U_1, because $Q^2 = 0$ in B, while $X_C \to X_A$ factors through U_2, because both W and W^{-1} are power-bounded in C and hence $|W| = 1$ on X_C.

The open subset U is proper. Use additive valuations with values in $\Gamma = \mathbf{Z}^3_{\text{lex}}$ and, for a nonzero restricted series written as in (6), put

$$v\left(\sum c_{a,b} W^a Q^b\right) = \min_{\text{lex}}\{(v_k(c_{a,b}), b, a) : c_{a,b} \neq 0\}.$$

Restrictedness makes the minimum exist, and the initial-term argument for the lexicographic monomial order makes v multiplicative. The condition $b \le 1 \Rightarrow a \ge 0$ defining S makes it nonnegative on $A_0 = A^\circ$, while $v(\varpi) > 0$, so it is continuous and defines a point $x \in X_A$. Since

$$v(Q^2) = (0,2,0) < (1,0,0) = v(\varpi), \qquad v(W) = (0,0,1) > 0,$$

one has $|Q^2(x)| > |\varpi(x)|$ and $|W(x)| < 1$; hence $x \notin U$.

Adic spaces may be glued along open subspaces. Glue two copies of X_A along U and write $Z = X_A^{(1)} \cup_U X_A^{(2)}$. The two open immersions $j_1, j_2 : X_A \to Z$ agree on U, so they agree after precomposition with both $X_B \to X_A$ and $X_C \to X_A$, but they are distinct at the two unglued copies of x. The displayed Hom map is therefore not injective for this Z. Thus X_A is not the pushout in the full category of adic spaces. More generally, the same doubling argument shows that a cocone whose two legs factor through a proper open subspace of its vertex cannot be a pushout in that category. We do not know whether the original span admits some other, necessarily nonaffinoid, pushout object.

Proof of theorem 1.1. Completeness, uniformity, nonnoetherianity, and the domain property are proposition 3.1; sheafiness is theorem 6.2; and failure of stable uniformity is proposition 4.2. □

7. A second example

The finite-jet construction above loses uniformity because separated completion creates a nilpotent. We now give a second construction in which the bad rational localisation is an integral domain. The loss of uniformity is instead caused by an unbounded family of nonnilpotent power-bounded elements.

Theorem 7.1. *There is a complete Tate k-algebra $\mathcal{A}$ with the following properties.*

1. *$\mathcal{A}$ is a uniform, nonnoetherian integral domain and $\mathcal{A}^\circ = \mathcal{A}_0$ for the ring of definition displayed below.*
2. *The Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.*
3. *Every finite iterated rational localisation of $\mathcal{A}$ is reduced.*
4. *The rational localisation*
$$\mathcal{B} = \mathcal{A}\left\langle \frac{W}{\varpi} \right\rangle$$
is an integral domain but is not uniform.

In particular, failure of stable uniformity need not be caused by a nilpotent in the bad rational localisation.

7.1. **The weighted-parity algebra.** Let $I = \mathbf{N}_{>0}$. For a finite-support multi-index $\nu = (\nu_n)_{n\in I} \in \mathbf{N}^{(I)}$, put

$$\omega(\nu) = \sum_{\substack{n \ge 1 \\ \nu_n \text{ odd}}} n. \tag{17}$$

Consider the submonoid

$$S = \left\{ (a,\nu) \in \mathbf{N} \times \mathbf{N}^{(I)} : a \ge \omega(\nu) \right\}. \tag{18}$$

The inequality

$$\omega(\nu + \mu) \le \omega(\nu) + \omega(\mu) \tag{19}$$

shows that S is closed under addition. It is locally finite: a fixed element of S has only finitely many decompositions as a sum of two elements of S.

Let $k[S]$ be the monoid algebra with basis $W^a U^\nu$, where $U^\nu = \prod_n U_n^{\nu_n}$. Give it the Gauss norm

$$\left\| \sum_{(a,\nu)\in S} c_{a,\nu} W^a U^\nu \right\|_G = \sup_{a,\nu} |c_{a,\nu}|. \tag{20}$$

Its completion is

$$\mathcal{A} = \left\{ \sum_{(a,\nu)\in S} c_{a,\nu} W^a U^\nu : c_{a,\nu} \in k, \quad c_{a,\nu} \longrightarrow 0 \right\}, \tag{21}$$

where convergence to zero is over the discrete index set S. Local finiteness makes the coefficientwise convolution finite, and the submultiplicativity of the Gauss norm on $k[S]$ extends multiplication continuously to $\mathcal{A}$. Put

$$\mathcal{A}_0 = \left\{ \sum_{(a,\nu)\in S} c_{a,\nu} W^a U^\nu \in \mathcal{A} : c_{a,\nu} \in k^\circ \right\}. \tag{22}$$

Then $\mathcal{A}_0$ is the closed unit ball for (20), it is ϖ-adically complete, and $\mathcal{A} = \mathcal{A}_0[1/\varpi]$.

The finite stages admit a useful affinoid presentation. For $N \geq 1$, put

$$\mathcal{A}_N = \frac{k\langle W, Y_1, Z_1, \ldots, Y_N, Z_N\rangle}{(Y_n^2 - W^{2n} Z_n)_{1\leq n\leq N}}. \tag{23}$$

Lemma 7.2 (Isometric finite-stage normal form)**.** *There is an isometric decomposition*

$$\mathcal{A}_N \cong \bigoplus_{\epsilon\in\{0,1\}^N}^{\max} k\langle W, Z_1, \ldots, Z_N\rangle Y^\epsilon. \tag{24}$$

Under the substitution

$$Y_n \longmapsto W^n U_n, \qquad Z_n \longmapsto U_n^2, \tag{25}$$

this identifies $\mathcal{A}_N$ isometrically with the closed subalgebra of $k\langle W, U_1, \ldots, U_N\rangle$ supported on the monomials in (18) *involving only $U_1, \ldots, U_N$. The transition maps $\mathcal{A}_N \to \mathcal{A}_{N+1}$ are isometric, and*

$$\mathcal{A} = \widehat{\bigcup_{N\geq 1} \mathcal{A}_N}. \tag{26}$$

Proof. Successive division by the monic relations $Y_n^2 - W^{2n} Z_n$ reduces every class uniquely to a sum

$$\sum_{\epsilon\in\{0,1\}^N} f_\epsilon(W, Z_1, \ldots, Z_N) Y^\epsilon.$$

The division process does not increase the Gauss norm, so the quotient norm is at most the maximum of the norms of the f_ϵ. Under (25), the summand indexed by ϵ has U_n-exponent congruent to ϵ_n modulo 2. Different parity classes therefore have disjoint monomial supports. The image of the normal form consequently has Gauss norm exactly $\max_\epsilon \|f_\epsilon\|$. Since the substitution is norm-nonincreasing on the original Tate algebra, the norm of this image is at most the quotient norm. This proves both uniqueness and the isometry (24).

Writing $\nu_n = 2q_n + \epsilon_n$ gives the unique factorization

$$W^a U^\nu = W^{a-\omega(\nu)} \prod_{n=1}^{N} Y_n^{\epsilon_n} Z_n^{q_n} \tag{27}$$

whenever $a \geq \omega(\nu)$ and ν is supported in $\{1, \ldots, N\}$. Thus the finite-stage image is exactly the claimed support algebra. The same description proves that the transition maps are isometric. Finally, every finite set of allowed monomials is contained in one finite stage, while finite-support series are dense in the restricted series (21); this proves (26). ☐

Proposition 7.3. *The ring $\mathcal{A}$ is a complete uniform Tate k-algebra and an integral domain, with*

$$\mathcal{A}^\circ = \mathcal{A}_0.$$

Proof. The Gauss norm on the countable restricted Tate algebra $k\langle W, U_1, U_2, \ldots\rangle$ is multiplicative. Indeed, every nonzero restricted series attains its norm. After scaling two nonzero series to norm one, reduction modulo ϖ leaves two nonzero polynomials in $\widetilde{k}[W, U_1, U_2, \ldots]$, and their product is nonzero. Scaling back proves multiplicativity. The support algebra $\mathcal{A}$ embeds isometrically into this Tate algebra by (21), so its Gauss norm is also multiplicative and it is a domain.

If $x \in \mathcal{A}_0$, all powers of x remain in $\mathcal{A}_0$. If $x \notin \mathcal{A}_0$, multiplicativity gives $\|x^m\|_G = \|x\|_G^m$, which is unbounded. Hence $\mathcal{A}^\circ = \mathcal{A}_0$. Completeness and the Tate property follow from (21) and the topologically nilpotent unit ϖ. ☐

Proposition 7.4. *The ring $\mathcal{A}$ is not noetherian.*

Proof. For $m \geq 1$, let

$$I_m = (Z_1, \ldots, Z_m) \subset \mathcal{A}.$$

There is a compatible family of norm-nonincreasing maps on the finite stages, and hence a bounded homomorphism

$$\psi_m : \mathcal{A} \longrightarrow k\langle T\rangle,$$

which sends W and every Y_j to zero, sends Z_{m+1} to T, and sends every other Z_j to zero. The relations in (23) are preserved. The map ψ_m kills I_m but not Z_{m+1}. Therefore

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \cdots,$$

so $\mathcal{A}$ is not noetherian. ☐

7.2. **A reduced nonuniform rational chart.** The ideal (W, ϖ) is open because ϖ is a unit of $\mathcal{A}$. Define the complete k°-algebra

$$\mathcal{B}_0 = \left\{ \sum_{\nu \in \mathbf{N}^{(I)},\ d \geq \omega(\nu)} b_{d,\nu} X^d U^\nu : b_{d,\nu} \in \varpi^{\omega(\nu)} k^\circ, \quad \varpi^{-\omega(\nu)} b_{d,\nu} \longrightarrow 0 \right\}, \tag{28}$$

with norm

$$\left\| \sum b_{d,\nu} X^d U^\nu \right\|_{\mathrm{in}} = \sup_{d,\nu} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|. \tag{29}$$

The subadditivity (19) shows that $\mathcal{B}_0$ is a ring. Put $\mathcal{B} = \mathcal{B}_0[1/\varpi]$.

Proposition 7.5. *There is a canonical isometric isomorphism*

$$\mathcal{A}\left\langle \frac{W}{\varpi} \right\rangle \xrightarrow{\sim} \mathcal{B}, \qquad W \longmapsto \varpi X. \tag{30}$$

The ideal $(W - \varpi X)$ in $\mathcal{A}\langle X\rangle$ is closed, so no additional separated quotient is required.

Proof. On rings of definition define

$$q_0 : \mathcal{A}_0\langle X\rangle \longrightarrow \mathcal{B}_0, \qquad q_0(W^a X^c U^\nu) = \varpi^a X^{a+c} U^\nu.$$

For a target monomial define a k°-linear section by

$$s\big(b_{d,\nu} X^d U^\nu\big) = \varpi^{-\omega(\nu)} b_{d,\nu} W^{\omega(\nu)} X^{d-\omega(\nu)} U^\nu. \tag{31}$$

The divisibility and support conditions in (28) make this well-defined, and (29) shows that s is isometric. Moreover, $q_0 s = \mathrm{id}$.

It remains to identify the kernel. Let $M = W^a X^c U^\nu$ and put $r = a - \omega(\nu) \geq 0$. If $r > 0$, set

$$D(M) = W^{\omega(\nu)} X^c U^\nu \sum_{j=0}^{r-1} W^{r-1-j} (\varpi X)^j,$$

and set $D(M) = 0$ when $r = 0$. Then

$$M - s q_0(M) = (W - \varpi X) D(M). \tag{32}$$

Each basis monomial occurring in $D(M)$ is still allowed in $\mathcal{A}_0\langle X\rangle$, and all its scalar coefficients have norm at most one. Thus D is norm-nonincreasing on finite-support series and extends continuously to the c_0-completion. If $x \in \ker(q_0)$, then

$$x = x - s q_0(x) = (W - \varpi X) D(x).$$

The reverse inclusion is immediate, so

$$\ker(q_0) = (W - \varpi X)\mathcal{A}_0\langle X\rangle. \tag{33}$$

Since q_0 is a split surjection onto the complete ring $\mathcal{B}_0$, this kernel is closed. The quotient norm is exactly (29): boundedness of q_0 gives one inequality, and the isometric section gives the reverse inequality. Inverting ϖ proves (30) and the closedness assertion. □

Proposition 7.6. *The ring $\mathcal{B}$ is an integral domain but is not uniform.*

Proof. The weighted convergence condition in (28) implies ordinary restricted convergence, because $|b_{d,\nu}| \leq |\varpi|^{\omega(\nu)} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|$. Hence coefficientwise inclusion gives an injective homomorphism

$$\mathcal{B} \hookrightarrow k\langle X, U_1, U_2, \ldots\rangle.$$

The target is a domain by the Gauss-norm argument used in proposition 7.3; therefore $\mathcal{B}$ is a domain.

Inside $\mathcal{A}$ write $Y_n = W^n U_n$, as in (25), and put

$$T_n = \varpi^{-n} Y_n = X^n U_n \in \mathcal{B}.$$

For $r \geq 0$, formula (29) gives

$$\|T_n^{2r}\|_{\mathrm{in}} = 1, \qquad \|T_n^{2r+1}\|_{\mathrm{in}} = |\varpi|^{-n}. \tag{34}$$

Thus every T_n is power-bounded. The family $(T_n)_{n\geq 1}$ is not bounded: if $\varpi^N T_n \in \mathcal{B}_0$, then the coefficient of $X^n U_n$ must be divisible by ϖ^n, which forces $N \geq n$. No fixed N works for all n. Hence $\mathcal{B}^\circ$ is unbounded and $\mathcal{B}$ is not uniform. □

7.3. **Small perturbations of rational data.** The finite-head argument rests on the following elementary stability result.

Lemma 7.7 (Small perturbation lemma)**.** *Let (E, E^+) be a complete Tate Huber pair and choose a closed ring of definition $E_0 \subseteq E^+$. Let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum whose entries lie in E_0. Write $d_0 = g$ and $d_i = f_i$ for $i \geq 1$. Suppose that for some $\ell \geq 0$ there are $a_0, \dots, a_m \in E_0$ with*

$$\varpi^\ell = \sum_{j=0}^{m} a_j d_j. \tag{35}$$

Let $d'_j \in E_0$ satisfy

$$d'_j - d_j \in \varpi^{\ell+1} E_0 \qquad (0 \leq j \leq m), \tag{36}$$

and write $\alpha' = (f'_1, \dots, f'_m; g')$. Then α' is a rational datum, α and α' define the same rational subset of $\mathrm{Spa}(E, E^+)$*, and their completed rational localisations are canonically isomorphic.*

Proof. Write $d'_j - d_j = \varpi^{\ell+1} h_j$ with $h_j \in E_0$. Then

$$\sum_j a_j d'_j = \varpi^\ell (1 + \varpi H), \qquad H = \sum_j a_j h_j \in E_0.$$

The element $1 + \varpi H$ is a unit of E_0, with inverse given by its convergent geometric series. Multiplying the a_j by this inverse gives an integral Bezout relation for the primed datum; in particular, α' is open.

Let x lie in the rational subset defined by α. Since $a_j \in E_0 \subseteq E^+$, relation (35) gives

$$|\varpi|^\ell \leq |g(x)|.$$

The perturbations have value at most $|\varpi|^{\ell+1} < |g(x)|$, so $|g'(x)| = |g(x)|$ and $|f'_i(x)| \leq |g'(x)|$. This proves one inclusion of rational subsets. Applying the same argument to the primed Bezout relation proves the reverse inclusion.

Let E_α be the first rational localisation and put

$$q = \frac{\varpi^\ell}{g} = a_0 + \sum_i a_i \frac{f_i}{g} \in E_\alpha^\circ.$$

Then

$$\frac{g'}{g} = 1 + \varpi h_0 q.$$

The second summand is topologically nilpotent. After adjoining the finitely many power-bounded elements involved to a ring of definition, the geometric series for its inverse and all powers of that inverse lie in the same bounded subring. Thus g'/g is a 1-unit with power-bounded inverse, and

$$\frac{f'_i}{g'} = \frac{f_i/g + \varpi h_i q}{1 + \varpi h_0 q}$$

is power-bounded. The universal property gives a homomorphism $E_{\alpha'} \to E_\alpha$. Repeating the argument with the primed Bezout relation gives a homomorphism in the opposite direction. Both composites restrict to the identity on E; universal-property uniqueness makes them inverse. The isomorphism is therefore canonical. $\square$

7.4. **Finite-head rational localisation.** Fix $N \geq 1$. For a tail multi-index $\mu = (\mu_n)_{n>N} \in \mathbf{N}^{(\{n>N\})}$, write $\mu_n = 2q_n + \epsilon_n$ with $\epsilon_n \in \{0,1\}$ and put

$$e_\mu = \prod_{n>N} Y_n^{\epsilon_n} Z_n^{q_n}. \tag{37}$$

The finite-stage normal forms give an isometric Banach-module decomposition

$$\mathcal{A} \cong \widehat{\bigoplus_\mu}^{c_0} \mathcal{A}_N e_\mu. \tag{38}$$

Multiplication is determined by

$$e_\mu e_\lambda = W^{\omega(\mu)+\omega(\lambda)-\omega(\mu+\lambda)} e_{\mu+\lambda}. \tag{39}$$

Here ω is applied only to the tail coordinates. Projection to the coefficient of e_0 is a norm-nonincreasing algebra retraction

$$\rho_N : \mathcal{A} \longrightarrow \mathcal{A}_N. \tag{40}$$

Proposition 7.8 (Coefficientwise localisation)**.** *Let α be a rational datum in $\mathcal{A}_N$, and put $P = (\mathcal{A}_N)_\alpha$. There is a canonical topological algebra isomorphism*

$$\mathcal{A}_\alpha \xrightarrow{\sim} \widehat{\bigoplus_\mu}^{c_0} P e_\mu. \tag{41}$$

It is natural for rational refinements represented in a finite head: if $P \to Q$ is the corresponding head restriction map, the induced restriction sends

$$\sum_\mu p_\mu e_\mu \longmapsto \sum_\mu (p_\mu|_Q) e_\mu. \tag{42}$$

The same statement holds after increasing N.

Proof. Let

$$R = \mathcal{A}_N\langle T_1, \dots, T_m\rangle, \qquad I = (gT_i - f_i)_i \subset R.$$

Since $\mathcal{A}_N$ is affinoid, the ideal I is closed and $d_1 : R^m \to I$ is strict; this is the degree-zero conclusion of lemma 5.1. In particular, there is a constant C such that every $y \in I$ has a lift $x \in R^m$ with $\|x\| \leq C\|y\|$.

By (38),

$$\mathcal{A}\langle T_1, \dots, T_m\rangle \cong \widehat{\bigoplus_\mu}^{c_0} R e_\mu.$$

Because the relations lie in the finite head, d_1 acts coefficientwise. Its image is exactly

$$\widehat{\bigoplus_\mu}^{c_0} I e_\mu.$$

Indeed, the inclusion from left to right is immediate. Conversely, for a null family $(y_\mu)_\mu$ in I, choose coefficientwise lifts x_μ with $\|x_\mu\| \leq C\|y_\mu\|$; then $(x_\mu)_\mu$ is again a null family. Thus the corresponding ideal over $\mathcal{A}$ is closed and

$$\frac{\widehat{\bigoplus}_\mu^{c_0} R e_\mu}{\widehat{\bigoplus}_\mu^{c_0} I e_\mu} \cong \widehat{\bigoplus_\mu}^{c_0} (R/I) e_\mu \tag{43}$$

as complete topological modules and as algebras. Since $R/I = P$, this proves (41).

Every map used in (43) is induced by the canonical quotient. If a further rational datum lies in the head, its first Koszul differential is again coefficientwise, so the same construction gives (42). Presentation independence and transitivity of rational localisation extend this compatibility to iterated refinements and to a larger finite head. ∎

Corollary 7.9 (Finite-head presentation)**.** *Every rational localisation of $\mathcal{A}$ is canonically isomorphic to a ring of the form*

$$\widehat{\bigoplus_{\mu}}^{c_0} Pe_\mu, \tag{44}$$

where P is a rational localisation of some affinoid head $\mathcal{A}_N$. Under this identification the inclusions obtained by enlarging the head have norm-nonincreasing algebra retractions, and their union is dense.

Proof. Start with a rational datum in $\mathcal{A}$. Multiplying all entries by one power of ϖ does not change the rational subset or localisation, so assume that the entries lie in $\mathcal{A}_0$. Choose an integral Bezout relation

$$\varpi^\ell = \sum_j a_j d_j, \qquad a_j \in \mathcal{A}_0.$$

By (26), for some N the entries d_j admit approximations $d'_j \in \mathcal{A}_N \cap \mathcal{A}_0$ satisfying $d'_j - d_j \in \varpi^{\ell+1}\mathcal{A}_0$. By lemma 7.7, the primed datum gives the same rational subset and the same completed localisation. The primed integral Bezout relation produced in that lemma remains valid after applying the retraction ρ_N; hence the primed datum is a rational datum already in $\mathcal{A}_N$. Now apply proposition 7.8.

For $M \geq N$, put $P_M = (\mathcal{A}_M)_\alpha$. The same proposition, with head M, identifies the localisation with the c_0-sum over the variables of index $> M$. Coefficient projection gives a norm-nonincreasing retraction to P_M, and finite tail truncation proves that $\bigcup_M P_M$ is dense. □

7.5. **Sheafiness.**

Theorem 7.10. *The Huber pair $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.*

Proof. We first prove the topological sheaf condition on the rational basis of $X = \operatorname{Spa}(\mathcal{A}, \mathcal{A}^\circ)$. Let $U \subset X$ be rational. By corollary 7.9, choose a head presentation

$$E = \mathcal{O}_X(U) \cong \widehat{\bigoplus_{\mu}}^{c_0} P_M e_\mu, \qquad P_M = (\mathcal{A}_M)_\alpha,$$

for some M. Let E_U^+ be the plus ring defining the rational subspace U; it need not equal the maximal ring E°. Choose the standard closed ring of definition $P_{M,0} \subset P_M^\circ$ furnished by the head graph presentation. The coefficientwise quotient by these relations gives a closed ring of definition

$$E_0 = \widehat{\bigoplus_{\mu}}^{c_0} P_{M,0} e_\mu \subseteq E_U^+.$$

The inclusion holds because E_U^+ is an open, hence closed, subring containing the integral graph algebra. After enlarging the head, finite tail truncation shows that the corresponding head lattices are dense in E_0.

Suppose that $U = \bigcup_{i=1}^r U_i$ is a finite rational cover. Choose rational data for the U_i inside the Huber pair (E, E_U^+). Scale their entries into E_0 and choose integral Bezout relations. After a common enlargement of the head, the density just noted and lemma 7.7 replace all these data by data lying in one head lattice $P_{M,0}$, without changing any U_i or its completed section ring. Applying the coefficient projection $E \to P_M$ to the integral Bezout relation supplied by lemma 7.7 shows that each new datum is rational already in P_M. Let

$$V_i \subset \operatorname{Spa}(P_M, P_M^\circ)$$

be the corresponding rational domains.

We verify that the V_i cover. The inclusion $P_M \to E$ and coefficient projection $E \to P_M$ are norm-nonincreasing and split each other. They therefore define morphisms of the maximal Huber pairs and a split surjection

$$\operatorname{Spa}(E, E^\circ) \longrightarrow \operatorname{Spa}(P_M, P_M^\circ). \tag{45}$$

The maximal-pair spectrum $\operatorname{Spa}(E, E^\circ)$ is a subspace of $\operatorname{Spa}(E, E_U^+) = U$. Hence the given cover restricts to a cover of $\operatorname{Spa}(E, E^\circ)$. On this subspace the restricted U_i are exactly the inverse images of the V_i. The surjectivity in (45) therefore implies that the V_i cover $\operatorname{Spa}(P_M, P_M^\circ)$. This avoids identifying U with the maximal-pair spectrum, which need not be correct at higher-rank points.

Write

$$P_{M,i} = \mathcal{O}(V_i), \qquad P_{M,ij} = \mathcal{O}(V_i \cap V_j).$$

Naturality in proposition 7.8 gives canonical identifications

$$\mathcal{O}_X(U_i) \cong \widehat{\bigoplus_\mu}^{c_0} P_{M,i} e_\mu, \qquad \mathcal{O}_X(U_i \cap U_j) \cong \widehat{\bigoplus_\mu}^{c_0} P_{M,ij} e_\mu, \tag{46}$$

under which every restriction map is coefficientwise.

The affinoid algebra P_M is sheafy, so its rational-cover equalizer is an isomorphism of Banach spaces

$$P_M \xrightarrow{\sim} \operatorname{Eq}\left(\prod_i P_{M,i} \rightrightarrows \prod_{i,j} P_{M,ij}\right). \tag{47}$$

Its inverse is bounded; fix a bound C. Let $(x_i)_i$ be a matching family in the section rings in (46), and write $x_i = \sum_\mu x_{i,\mu} e_\mu$. For each μ, equation (47) glues $(x_{i,\mu})_i$ to a unique $x_\mu \in P_M$, with

$$\|x_\mu\| \le C \max_i \|x_{i,\mu}\|. \tag{48}$$

Because the cover is finite and every coefficient family is null, the right side tends to zero. Thus $\sum_\mu x_\mu e_\mu$ is an actual element of the c_0-sum and gives the unique global section. The same bound proves that the restriction map is a topological embedding. Hence the structure presheaf satisfies the sheaf condition as complete topological rings on the rational basis.

For an arbitrary open cover, apply the projective-limit argument used in the last paragraph of the proof of lemma 6.1. Namely, every rational subdomain of the open set is quasi-compact and admits a finite rational cover subordinate to the given cover; the rational-basis result glues continuously on that subdomain, and uniqueness makes these sections compatible under rational restriction. The defining projective-limit topology on arbitrary-open sections then makes the inverse gluing map continuous. Thus the public all-open structure presheaf is a sheaf of complete topological rings.

□

7.6. **Reducedness of all rational localisations.**

Lemma 7.11. *Let P be a reduced ring and let J be any index set. Define*

$$\mathcal{F}_J(P) = \prod_{\mu \in \mathbf{N}^{(J)}} P\, U^\mu, \tag{49}$$

with multiplication by coefficientwise finite convolution. Then $\mathcal{F}_J(P)$ is reduced.

Proof. For every finite subset $J_0 \subset J$, setting the variables outside J_0 equal to zero gives a homomorphism

$$\tau_{J_0} : \mathcal{F}_J(P) \longrightarrow P[[U_j : j \in J_0]].$$

These homomorphisms jointly detect all coefficients. A finite-variable formal power-series ring over a reduced ring is reduced: the injection $P \to \prod_{\mathfrak{p} \in \mathrm{Spec}(P)} P/\mathfrak{p}$ induces an injection of formal power-series rings into a product of domains. If an element of $\mathcal{F}_J(P)$ is nilpotent, every τ_{J_0} kills it; joint coefficient detection then kills the original element. □

Theorem 7.12. *Every finite iterated rational localisation of $\mathcal{A}$ is reduced.*

Proof. By transitivity of rational localisation, it is enough to treat one rational localisation. Use corollary 7.9 to write it as

$$E \cong \widehat{\bigoplus_{\mu}}^{c_0} P e_\mu, \qquad P = (\mathcal{A}_N)_\alpha. \tag{50}$$

The affinoid algebra $\mathcal{A}_N$ is a domain by lemma 7.2. A rational localisation of a reduced affinoid algebra is reduced [4, Corollary 7.3.2/10]; see also [9, Remark 1.2.16]. Thus P is reduced.

The element W is a non-zero-divisor in $\mathcal{A}_N$. Affinoid rational localisation is algebraically flat [4, Corollary 7.3.2/6]; see also [10, Proposition 15.1.1, proof]. Tensoring $0 \to \mathcal{A}_N \xrightarrow{W} \mathcal{A}_N$ with P therefore shows that multiplication by W is injective on P. If $P = 0$, the conclusion is immediate, so assume otherwise.

Let $J = \{n > N\}$. Define

$$\Phi : E \longrightarrow \mathcal{F}_J(P), \qquad \Phi\left(\sum_\mu x_\mu e_\mu\right) = \sum_\mu W^{\omega(\mu)} x_\mu U^\mu. \tag{51}$$

The multiplication rule (39) makes Φ multiplicative. If $\Phi(\sum x_\mu e_\mu) = 0$, coefficient comparison gives $W^{\omega(\mu)} x_\mu = 0$ for every μ. The W-regularity of P then gives $x_\mu = 0$ for every μ, so Φ is injective. By lemma 7.11, the target is reduced; hence E is reduced. □

Proof of theorem 7.1. The uniform domain assertion is proposition 7.3, and nonnoetherianity is proposition 7.4. Sheafiness is theorem 7.10. Reducedness of all finite iterated rational localisations is theorem 7.12. Finally, propositions 7.5 and 7.6 identify the rational chart $|W| \le |\varpi|$ as a nonuniform integral domain. □

Remark 7.13 (Relation with the first construction)**.** The finite-jet and weighted-parity examples use different mechanisms. In the finite-jet ring, a whole tail becomes infinitely ϖ-divisible and the bad chart acquires a nilpotent. In the weighted-parity ring, every rational calculation can be moved into a noetherian finite head and then carried coefficientwise through a c_0-tail. Its bad chart remains a domain, and nonuniformity is witnessed by the unbounded family $(T_n)_{n \ge 1}$ of nonnilpotent power-bounded elements. The use of an infinite monomial support and of unbounded power-bounded directions is in the tradition of the examples of Buzzard–Verberkmoes and Mihara; the finite-head coefficientwise descent is the additional feature that makes sheafiness and reducedness after iterated rational localisation accessible here.

Appendix A. Lean formalisation and verification

This appendix records the formalisation of the finite-jet construction and the present state of the weighted-parity construction. The finite-jet source is AINTLIB commit `b007a4f3d`; the weighted-parity source is the later work-in-progress commit `090a28921` on the branch `wp/reduced-example` [1]. The latter contains the construction and proves uniformity, sheafiness, nonnoetherianity, the domain property, and failure of stable uniformity. It does not yet prove, without unproved inputs, that every finite iterated rational localisation is reduced.

Neither pinned snapshot was available from the public AINTLIB repository at the time of writing. The interactive version of the paper therefore includes each declaration used below, extracted from the appropriate snapshot. The verification report records the Lean and mathlib versions, the build, and the axiom checks.

A.1. **Sheafiness.** The finite-jet proof first establishes a condition on finite rational covers. Its historical internal class is `IsSheafy`, in the `ValuationSpectrum` namespace and defined in `StructureSheaf.lean`.

```
class IsSheafy (A : Type u) [CommRing A] [TopologicalSpace A]
    [IsTopologicalRing A] [inst₁ : PlusSubring A]
    [inst₂ : IsHuberRing A] [T2Space A] [NonarchimedeanRing A]
    [letI : UniformSpace A :=
      IsTopologicalAddGroup.rightUniformSpace A; CompleteSpace A]
    [IsRingOfIntegralElements (A⁺)] : Prop where
  embedding : ∀ (C : RationalCoveringData A), C.IsRational →
    Topology.IsEmbedding (productRestrictionSub A C)
  gluing : ∀ (C : RationalCoveringData A), C.IsRational →
    ∀ (f : ∀ (D : ↑C.covers), presheafValue D.1),
    (∀ (D₁ D₂ : ↑C.covers) (D₃ : RationalLocData A)
      (h₃₁ : rationalOpen D₃.T D₃.s ⊆
        rationalOpen D₁.1.T D₁.1.s)
      (h₃₂ : rationalOpen D₃.T D₃.s ⊆
        rationalOpen D₂.1.T D₂.1.s),
      restrictionMap D₁.1 D₃ h₃₁ (f D₁) =
        restrictionMap D₂.1 D₃ h₃₂ (f D₂)) →
    ∃ x : presheafValue C.base, ∀ (D : ↑C.covers),
      restrictionMap C.base D.1 (C.hsubset D.1 D.2) x = f D
```

The two fields say that restriction to a finite rational cover is a topological embedding and that every matching family glues. Compatibility is written using common rational refinements; the exact-intersection theorem identifies this with the usual compatibility on pairwise intersections. Thus `IsSheafy A` is the topological sheaf condition on the rational basis.

For an arbitrary open subset $V \subseteq \mathrm{Spa}(A, A^+)$, `limitSections V` is the projective limit of the sections over the rational subdomains contained in V. The corresponding all-open condition is `IsLimitSheaf`, from `SheafyPair.lean`.

```
structure IsLimitSheaf (A : Type u) [CommRing A] [TopologicalSpace A]
    [IsTopologicalRing A] [PlusSubring A] [IsHuberRing A]
    [HasLocLiftPowerBounded A] : Prop where
  injective : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺)))
    {x y : ↑(limitSections V)},
    (∀ i, limitRestrict (hle i) x = limitRestrict (hle i) y) → x = y
  glue : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺)))
    (s : ∀ i, ↑(limitSections (U i))),
    (∀ i j,
      limitRestrict (inf_le_left (a := U i) (b := U j)) (s i) =
```

```
      limitRestrict (inf_le_right (a := U i) (b := U j)) (s j)) →
    ∃ x : ↑(limitSections V), ∀ i, limitRestrict (hle i) x = s i
  isEmbedding : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺))),
    Topology.IsEmbedding (limitRestrictProd hle)
```

The fields `injective` and `glue` are the usual sheaf axioms on arbitrary opens. The final field says that the resulting map to the matching-family equaliser is also a topological embedding. This is the extra condition needed for a sheaf of complete topological rings.

For complete Tate pairs the rational-basis and all-open predicates are equivalent:

```
theorem isSheafy_iff_isLimitSheaf :
    IsSheafy A ↔ IsLimitSheaf A :=
  ⟨fun _ => isLimitSheaf_of_isSheafy,
    isSheafy_of_isLimitSheaf⟩
```

The forward implication is the sheaf-on-a-rational-basis theorem, and the converse is restriction to rational covers. The structure-presheaf equivalence identifies `IsLimitSheaf A` with Wedhorn's topological-ring formulation [18, Remark 8.20]. Mathlib's ordinary categorical sheaf predicate is weaker because it does not record the topology.

For a complete Tate ring the following definitions fix a ring of integral elements and then remove the choices of plus ring and completion. They are in `SheafyRing.lean`.

```
def IsSheafyFor (Aplus : RingOfIntegralElements A) : Prop :=
  letI := Aplus.toPlusSubring
  haveI : IsRingOfIntegralElements (A⁺ : Subring A) := Aplus.2
  haveI : HasLocLiftPowerBounded A := hasLocLiftPowerBounded_faithful
  IsLimitSheaf A

def IsSheafyComplete : Prop :=
  ∀ Aplus : RingOfIntegralElements A, IsSheafyFor A Aplus

def IsSheafyTateRing : Prop :=
  ∀ (P : PairOfDefinition A)
    (Bplus : RingOfIntegralElements (CompletionModel A P)),
    IsSheafyFor (CompletionModel A P) Bplus
```

The predicate `IsSheafyFor A Aplus` says that the Huber pair (A, A^+) is sheafy. For complete Tate rings, the chosen-plus-ring equivalence shows that this is independent of the chosen ring of integral elements. The equivalences in `SheafyCompletionModel.lean` remove the choice of completion. Consequently `IsSheafyTateRing` is the analytic specialisation of Wedhorn's Definition 8.26 [18, Definition 8.26].

A.2. **Strong noetherianity.** The definitions below are from `RestrictedPowerSeries.lean`.

```
def MvPowerSeries.IsRestricted {k : ℕ} {A : Type*}
    [CommRing A] [TopologicalSpace A]
    (f : MvPowerSeries (Fin k) A) : Prop :=
  Tendsto (fun s : Fin k →₀ ℕ => MvPowerSeries.coeff s f)
    cofinite (nhds 0)

class IsStronglyNoetherian (A : Type*) [CommRing A]
    [TopologicalSpace A] [NonarchimedeanRing A] : Prop where
```

```
  isNoetherianRing_restricted : ∀ k : ℕ,
    IsNoetherianRing (restrictedMvPowerSeriesSubring k A)
```

The cofinite convergence condition is the usual condition that the coefficients tend to zero, so `restrictedMvPowerSeriesSubring k A` represents $A\langle T_1, \ldots, T_k\rangle$. The second declaration therefore says that these rings are noetherian for every k. Since A is already complete, this is Wedhorn's definition of strong noetherianity [18, Proposition and Definition 6.36].

A.3. **The finite-jet example.** We finally record the definition of the example itself. In the following code, `PowerSeries.Restricted R 1` is the radius-one Tate algebra in one variable over R, `RestrictedLaurent K` is $K\langle W, W^{-1}\rangle$, and `DualNumber R` is $R[Q]/(Q^2)$. The source is `FJP/Over/JetRings.lean`.

```
abbrev L : Type u := RestrictedLaurent K

abbrev JetC : Type u := PowerSeries.Restricted (L K) (1 : ℝ)

abbrev JetB : Type u :=
  DualNumber (PowerSeries.Restricted K (1 : ℝ))

abbrev JetD : Type u := DualNumber (L K)

noncomputable def jetSupport : Subring (JetC K) where
  carrier := {f |
    qCoeff K 0 f ∈ nonnegSubring K ∧
    qCoeff K 1 f ∈ nonnegSubring K}
  zero_mem' := by
    constructor <;> rw [qCoeff_zero] <;> exact zero_mem _
  one_mem' := by
    constructor <;> rw [qCoeff_one]
    · rw [if_pos rfl]; exact one_mem _
    · rw [if_neg one_ne_zero]; exact zero_mem _
  add_mem' := fun {f g} hf hg => by
    constructor <;> rw [qCoeff_add]
    · exact add_mem hf.1 hg.1
    · exact add_mem hf.2 hg.2
  neg_mem' := fun {f} hf => by
    constructor <;> rw [qCoeff_neg]
    · exact neg_mem hf.1
    · exact neg_mem hf.2
  mul_mem' := fun {f g} hf hg => by
    constructor
    · rw [qCoeff_zero_mul]
      exact mul_mem hf.1 hg.1
    · rw [qCoeff_one_mul]
      exact add_mem (mul_mem hf.1 hg.2) (mul_mem hf.2 hg.1)

abbrev JetA : Type u := ↑(jetSupport K)
```

The predicate `nonnegSubring K` cuts out $K\langle W\rangle$ inside $K\langle W, W^{-1}\rangle$. Consequently the carrier condition says that the coefficients of Q^0 and Q^1 have no negative powers of W, while the coefficients of Q^n for $n \geq 2$ are unrestricted. Thus `JetA K` is

$$\left\{ f_0(W) + Qf_1(W) + Q^2 h(W, W^{-1}, Q) \;\middle|\; \substack{f_0, f_1 \in K\langle W\rangle \\ h \in L\langle Q\rangle} \right\},$$

which is the coefficient description of A in equation (5) after inverting ϖ.

The file also constructs the four maps in the Milnor square and proves `milnorRow_exact`: every compatible pair in `JetB K` and `JetC K` comes from a unique element of `JetA K`. This identifies `JetA K` with $B \times_D C$. The sheaf argument produces the finite-rational-cover condition displayed above:

```
include ϖ hK₀ in
theorem isSheafy_JetA :
    ValuationSpectrum.IsSheafy (JetA K) where
  embedding := fun C hC =>
    productRestrictionSub_isEmbedding_JetA ϖ hK₀ C hC
  gluing := fun C hC f hcompat =>
    gluing_JetA ϖ hK₀ C hC f hcompat
```

Here ϖ is a `Uniformizer K`, and the remaining input is the noetherianity of `unitBall K`. This is the version of sheafiness proved directly for the example. The equivalences above give the all-open and choice-independent statements. The main theorems are collected in `SheafyEndpoints`, and include the following.

```
theorem finiteJet_isDomain : IsDomain (JetA K) :=
  inferInstance

theorem finiteJet_not_noetherian :
    ¬ IsNoetherianRing (JetA K) :=
  not_isNoetherianRing_JetA K

theorem finiteJet_isUniform_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    TopologicalRing.IsUniform (JetA K) :=
  finiteJet_isUniform K (Uniformizer.ofDVR K)

theorem finiteJet_not_stablyUniform_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    ¬ TopologicalRing.IsStablyUniform (JetA K) :=
  finiteJet_not_stablyUniform K (Uniformizer.ofDVR K)

theorem finiteJet_isSheafyFor_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    IsSheafyFor (JetA K) (finiteJetPlus K) :=
  finiteJet_isSheafyFor K (Uniformizer.ofDVR K)
    (isNoetherianRing_unitBall K)

theorem finiteJet_isSheafyTateRing_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    IsSheafyTateRing (JetA K) :=
  finiteJet_isSheafyTateRing K (Uniformizer.ofDVR K)
    (isNoetherianRing_unitBall K)
```

Here K is any complete nontrivially normed ultrametric field for which $\mathcal{O}[K] = \{x : \|x\| \leq 1\}$ is a discrete valuation ring. No local compactness, finiteness of the residue field, characteristic, or perfection assumption is used. The theorem `finiteJet_isSheafyFor_of_dvr` proves that (A, A°) is sheafy, whilst `finiteJet_isSheafyTateRing_of_dvr` removes the choices of completion and ring of integral elements. Despite its name, the latter does not quantify over finite Tate extensions of A, so it is not a strong sheafiness statement. The corresponding all-open statement is

```
TopCat.Presheaf.IsSheafOfTopologicalRings
    (ValuationSpectrum.structurePresheaf (JetA K)),
```

The corresponding Lean theorem proves this statement directly.

A.4. **Scope of the finite-jet formalisation.** The paper's A_0 is represented by the Gauss-norm unit ball. The theorem `isPowerBounded_JetA_iff` identifies this unit ball with the power-bounded elements, giving $A^\circ = A_0$. Completeness and the Tate structure are supplied by the corresponding instances.

For the bad chart, `chartEquiv` is an exported ring equivalence. The two continuity theorems make it a homeomorphism, and the canonical-map theorem identifies W and Q. It is not bundled as a `K-AlgEquiv`, which is why proposition 4.1 is stated for complete topological rings.

The conclusions of lemma 5.1 are formalised in the normed setting used for the example: T_E has its Gauss norm, $T_{E,0}$ is its unit ball, and the finite free terms have their sup norms. The statement for an arbitrary noetherian ϖ-adic E_0 follows from Huber's finite-module results and comparison of lattices, but this comparison is not separately bundled in Lean. Similarly, the development proves the finite-jet instance `isSheafy_JetA` of lemma 6.1, rather than the abstract lemma for every strict Milnor square. The components of proposition 5.3 are separate declarations.

The Koszul argument, the rational-chart calculation, and the transfer of sheafiness through the concrete Milnor square are contained respectively in the restricted-exactness file and the strictness and closed-image file, `FJP/Over/Chart.lean`, and `FJP/Over/SheafTransfer.lean`.

A.5. **The weighted-parity example.** The second development uses the same restricted-series infrastructure. Its ambient ring is the countable restricted Tate algebra $K\langle W, U_1, U_2, \ldots\rangle$. The definitions `wpWeight`, `WPMem`, and `wpSupport` are the direct translation of (17)–(21). The relevant part of the definition is as follows.

```
def wpWeight (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : ℕ :=
  ∑ n ∈ t.support,
    if (t n).mod 2 = 1 ∧ n ≠ 0 then w n else 0

def WPMem (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : Prop :=
  wpWeight w t ≤ t 0

abbrev WPA : Type _ := ↑(wpSupport K w)

def idWeight : ℕ → ℕ := fun n => n

abbrev WPAid : Type _ := WPA K idWeight
```

The coordinate $t(0)$ is the exponent of W; the other coordinates are the exponents of the U_n. Thus `WPMem idWeight t` says exactly that $a \geq \omega(\nu)$, and `wpSupport` requires the coefficients outside this monoid to vanish. It follows that `WPAid K` is the ring $\mathcal{A}$ of (21). The maps `constA` and `norm_constA` give the isometric embedding of K as the constant series. This map is explicit in the development, although it is not bundled as an `Algebra K` instance.

Suppose that K is complete and nontrivially normed and that $\mathcal{O}[K]$ is a DVR. The principal exported statements are:

```
theorem weightedParity_isUniform_of_dvr :
    IsUniform (WPAid K) :=
  weightedParity_isUniform (Uniformizer.ofDVR K)
```

```
theorem weightedParity_isDomain :
    IsDomain (WPAid K) :=
  inferInstance

theorem weightedParity_not_noetherian :
    ¬ IsNoetherianRing (WPAid K) :=
  not_isNoetherianRing_WPA K idWeight idWeight_pos

theorem weightedParity_powerBounded_eq_unitBall
    (ϖ : Uniformizer K) :
    powerBoundedSubring (WPAid K) =
      (FiniteJet.unitBall (WPAid K) : Set (WPAid K)) :=
  powerBoundedSubring_eq_unitBall ϖ

theorem weightedParity_isSheafyComplete_of_dvr :
    IsSheafyComplete (WPAid K) :=
  weightedParity_isSheafyComplete (Uniformizer.ofDVR K)
    (FiniteJetOver.isNoetherianRing_unitBall K)

theorem weightedParity_chart_isDomain
    (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
    IsDomain (presheafValue
      (chartDatum (w := idWeight) ϖ)) :=
  isDomain_chart ϖ hK₀

theorem weightedParity_chart_not_isUniform
    (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
    ¬ IsUniform (presheafValue
      (chartDatum (w := idWeight) ϖ)) :=
  not_isUniform_chart idWeight_unbounded ϖ hK₀

theorem weightedParity_not_stablyUniform_of_dvr :
    ¬ IsStablyUniform (WPAid K) :=
  weightedParity_not_stablyUniform (Uniformizer.ofDVR K)
    (FiniteJetOver.isNoetherianRing_unitBall K)
```

Here `IsSheafyComplete` is the predicate defined above: it quantifies over every ring of integral elements. In particular it proves the sheaf condition for $(\mathcal{A}, \mathcal{A}^\circ)$. The all-open theorem also records the corresponding statement directly for the structure presheaf. The datum `chartDatum` is $(W; \varpi)$, and `chartDatum_isRational` checks that it defines the chart $|W| \le |\varpi|$. The two chart theorems then say precisely that its section ring is a domain but is not uniform.

These declarations, including the final failure of stable uniformity, have been checked to use only the usual logical axioms `propext`, `Classical.choice`, and `Quot.sound`. The current declarations asserting reducedness after every finite chain of rational localisations, on the other hand, still depend on `sorryAx`. Clause (3) of theorem 7.1 is therefore not counted as formalised. The branch also proves sheafiness for the shifted-weight models and constructs bicontinuous equivalences with the corresponding Tate extensions, but does not yet package the transported statement as a single strong-sheafiness theorem; no such claim is made here.

A.6. **Mathematical and code provenance.** The formal structure presheaf, rational-cover criterion, and the strongly-noetherian sheaf theorem follow Huber's construction [8, Theorem 2.2

and Lemmas 2.3–2.6] and Wedhorn's systematic formulation [18, Proposition and Definition 6.36, Remark 8.20, Definition 8.26, Theorem 8.28(b), Corollary 8.32, and Lemmas 8.33–8.34]. The passage between a fixed plus ring and all plus rings follows the standard rational-refinement argument recorded by Kedlaya [9, Lemma 1.6.8 and Remark 1.6.9]. The strict Koszul input is due to Ben-Bassat–Kremnizer and Bambozzi–Kremnizer; Huber's finite-module lemmas and the bounded-denominator criterion of Buzzard–Verberkmoes supply the lattice form used in the proof.

The development is written in Lean 4 [12] on top of mathlib [13]. The formalisation-specific files live in AINTLIB [1]. Its restricted power-series and restricted Laurent-series infrastructure includes Apache-2.0-licensed code adapted from William Coram's `PhD` repository, with equivalence code attributed there to Bingyu Xia [6]. The vendored source headers and commit history retain those attributions. The finite-jet and weighted-parity definitions, localisation proofs, and sheaf endpoints discussed in this appendix are the AINTLIB-specific part of the development.

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